

Static Electricity

The Dancing Cereal Experiment

Session 1: The Invisible Pull

Are You a Science Magician?

Have you ever taken off your school sweater on a cold winter morning and felt a tiny 'tick-tick' sound? Or watched your hair try to fly away while combing? Today, we will learn how to control this invisible power to make cereal dance!

Your Science Mission

In this session, we will put on our magician's hats to achieve these goals:

- **Understand Static Electricity:** Discover the "sticky power" made by rubbing. [cite: 1]
- **Master the Magic of Rubbing:** Learn how surfaces trade invisible charges. [cite: 1]
- **The Dancing Cereal Demo:** Use a balloon to make *Muri* (puffed rice) jump! [cite: 1]
- **Find Static at Home:** Spot everyday examples in winter clothes and blankets. [cite: 1]

THE QUICK MYSTERY

Can you make hair stand up without touching it? Let's find out how!

What is Static Electricity?

Most electricity flows through wires to power our lights and fans. But **Static Electricity** is different—it likes to stay in one place on the surface of an object! [cite: 1]

Definition: Static Electricity is a build-up of electrical charge on the surface of an object, usually made by rubbing two things together. [cite: 1]

The Magic of Rubbing

Everything in the world is made of tiny invisible pieces. When we rub two things together (like a balloon and a woollen sweater), they trade these pieces. This makes the balloon "charged" and ready for magic! [cite: 1]

[Visual: Illustration of a balloon being rubbed against a woollen school sweater]

Static Electricity in India

You don't need a lab to see static electricity. It is happening all around us in our Indian homes! [cite: 1]

1. The Winter Sweater

When you take off your school sweater during a cold morning assembly, you might hear tiny 'zaps'. That's static electricity escaping! [cite: 1]

2. The Silk Saree Spark

Have you seen your mother's silk saree "stick" to her or even seen a tiny spark in the dark? Silk is a great material for building up charge. [cite: 1]

3. The Plastic Comb

After a bath, when you dry your hair and use a plastic comb, sometimes your hair tries to follow the comb. The comb has stolen some charge! [cite: 1]

Winter Fact

Static electricity is much stronger in dry winter air than in the humid rainy season! [cite: 1]

The Experiment: Dancing Cereal

It's time for our main act! We are going to make *Muri* (puffed rice) dance without touching it. [cite: 1]

What You Need:

- A handful of Puffed Rice (*Muri* or *Pori*). [cite: 1]
- A balloon (bright colours are best!). [cite: 1]
- A woollen cloth or your own dry hair. [cite: 1]
- A flat plate. [cite: 1]

SCIENCE PREDICTION

What do you think will happen when the balloon gets close to the cereal? Will it pull it or push it away? [cite: 1]

Making the Magic Happen

Follow these steps carefully to start the dance: [cite: 1]

Step 1: Spread the Dancers

Spread a few pieces of *Muri* on a flat plate. Make sure they are dry. [cite: 1]

Step 2: Charge the Wand

Blow up the balloon. Rub it back and forth very quickly against your dry hair or a woollen sweater for 30 seconds. You are building up "Sticky Power"! [cite: 1]

Step 3: The Performance

Slowly bring the balloon close to the *Muri*. Do not touch them! [cite: 1]

[Visual: Puffed rice jumping up and 'dancing' underneath the balloon]

The Science Secret

Why did the *Muri* jump up? It's not real magic—it's physics! [cite: 1]

The Invisible Pull

When you rubbed the balloon, it became full of static electricity. This gave it an **Invisible Pull** (Attraction). Because *Muri* is very light, the balloon's pull is strong enough to lift it into the air! [cite: 1]

The Jump and Fall

Sometimes the *Muri* hits the balloon and then flies away. This is because it "steals" some charge from the balloon and then gets pushed away! [cite: 1]

Static Electricity Challenges

Try these other "magic" tricks at home to see where else static electricity is hiding: [cite: 1]

1. THE BENDING WATER TRICK

Turn on a tap so only a very thin stream of water comes out. Rub your plastic comb through your hair and hold it near the water. Watch the water bend! [cite: 1]

2. THE PAPER PICKER

Tear up tiny bits of tissue paper. Charge your balloon and see how many bits you can pick up at once. [cite: 1]

3. THE BALLOON WALL-STICK

Charge your balloon and try to stick it to the wall. How long does it stay there before the static charge wears off? [cite: 1]

The Science Safety Guide

Static electricity is fun, but we must use it wisely. [cite: 1]

SAFETY FIRST!

- **Tiny Shocks:** Static sparks might give you a tiny surprise (like a pin-prick), but they are not dangerous. However, never play with real electrical sockets or wires! [cite: 1]
- **Balloon Care:** Don't rub the balloon so hard that it pops. Pop sounds can be scary for pets! [cite: 1]
- **Clean Up:** Always put your *Muri* back in the bin or give it to the birds after your experiment. [cite: 1]

Lesson Recap

You have successfully completed your first session as a Science Magician! Let's remember the big ideas: [cite: 1]

- **Static Electricity** is built-up charge on a surface. [cite: 1]
- **Rubbing** is the key to creating this "Sticky Power". [cite: 1]
- It can **Pull** light objects like cereal or paper without touching them. [cite: 1]
- We see it in **sweaters, combs, and silk sarees**. [cite: 1]

Stay Curious!

The world is full of invisible forces waiting for you to find them. See you in the next session! [cite: 1]